



KISANKAVACH

Next-Generation Digital Public Infrastructure for Agricultural Credit & Fraud Defense

State Level Bankers' Committee (SLBC) Karnataka — Anchor Presentation

Anchor Convener:	Canara Bank (SLBC Convener) & Member Banks (SBI, Karnataka Gramin Bank, Karnataka Bank)
Government & Regulators:	Dept. of Agriculture, Bhoomi (Revenue), FRUITS, CeG, Reserve Bank of India (RBI), NABARD
Presenter & Author:	Sarath Babu Rayaprolu (Founder & Solution Architect)
Live Platform & APK:	https://kisan.digikavach.net Native Android APK Build Ready
Meeting Objective:	Formal Empanelment, Core Banking System (CBS) API Authorization & Statewide Rollout across 31 Districts

The Agrarian Credit Bottleneck in Karnataka

India's 140M farmers and Karnataka's 8.2M landholders face deep operational friction in institutional credit delivery, creating severe costs for banks and driving farmers to moneylenders.

FARMER CHALLENGES	BANK OPERATIONAL FRICTION	KISANKAVACH TRANSFORMATION
• 30 to 45 Day Turnaround: Farmers wait over a month for basic KCC credit. • Rs. 3,500+ Lost per Loan: Travel, wage losses & private documentation agents. • 24%-36% Moneylender Trap: High borrowing costs due to bank delays. • Zero Visibility: Opacity leads to repeated physical branch visits. • Cyber Scams: Farmers victimized by WhatsApp/SMS OTP phishing.	• Rs. 2,200+ Cost per File: High physical processing and manual survey costs. • 38% Rejection Rate: Disconnected, outdated, or defective land titles. • Ghost Borrowers & Fraud: Impersonation and forged physical land records. • Multi-Bank Lien Risk: Same land mortgaged across multiple institutions. • PSL Pressure: High compliance stress meeting RBI Priority Sector quotas.	✓ Turnaround < 7 Days: 80% acceleration via instant digital validation. ✓ Rs. 1,500 Saved per Loan: Origination cost reduced from Rs. 2,200 to Rs. 700. ✓ 100% Verified Identity: Direct Bhoomi RTC & FRUITS crop integration. ✓ Zero Ghost Borrowers: Aadhaar-locked biometric and title provenance. ✓ Proactive Cyber Shield: Heuristic phishing detection protecting accounts.

System Architecture: Zero Statutory Displacement DPI

Architectural Mandate: KisanKavach does NOT replace statutory systems of record (Bhoomi or FRUITS). Instead, it serves as an enterprise orchestration and fraud-defense highway connecting state registries directly into Bank Core Banking Systems (CBS).

1. BHOOMI REGISTRY	2. FRUITS PLATFORM	3. BANK LOS & CBS	4. CYBERKAVACH
Land Titles & Encumbrance	Farmer Identity & Crops	Maker-Checker Portal	AI Fraud & Scam Defense
• Real-time RTC / Pahani API query • Survey No. & Hissa validation • Exact extent & soil classification • Sub-registrar encumbrance check • Automated bank charge creation	• Direct Farmer ID (FID) verification • Seasonal crop declaration history • Aadhaar-linked DBT eligibility • Scale of Finance validation • Zero duplicate benefit leakage	• Auto-prefilled loan applications • Instant CIBIL bureau credit pull • Maker-Checker underwriting queue • Mandatory rejection taxonomy • Core Banking ISO-8583 / REST push	• Rural phishing & SMS heuristic scan • Predatory loan APK detector • Kannada & English scam parsing • Real-time fraud advisory push • Disbursement account protection

10-District Karnataka Pilot: Quantified Performance

Live operational metrics across 10 diverse pilot districts validating speed, scale, and loss elimination.

FARMERS ONBOARDED 1,50,000+ Across 10 Pilot Districts	KCC LOANS SANCTIONED 1,05,000 70.0% Sanction Conversion	CREDIT FACILITATED Rs. 1,890 Cr Avg Rs. 1.80L per sanction	AVERAGE TURNAROUND 6.2 Days Down from 35 days (82% faster)
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PILOT DISTRICT DISTRIBUTION	KEY BANKING CONSORTIUM OUTCOMES
• Mysuru: 16,400 Sanctions (Southern Agri Belt) • Mandya: 14,800 Sanctions (Sugarcane & Paddy) • Belagavi: 15,300 Sanctions (Commercial & Cotton) • Hassan: 11,200 Sanctions (Plantation Crops) • Kalaburagi: 10,100 Sanctions (Pulses & Tur Hub) • Ballari, Tumakuru, Shivamogga, Davanagere, Bagalkote: 37,200 Sanctions	✓ Zero Ghost Borrowers: Zero counterfeit land records or ghost claims. ✓ Application Accuracy: Incomplete files dropped from 38% to under 2.1%. ✓ Field Appraisals: Field officers executed 4.2x more appraisals per day. ✓ Branch Operational Savings: Participating branches saved ~Rs. 15.75L each. ✓ Farmer Satisfaction: 94.2% satisfaction index across district centers.

Underwriting Revolution: Bank Maker-Checker Portal

Standardizing credit underwriting with zero paperwork, instant bureau scores, and automated SLA compliance.

STAGE 1: MAKER APPRAISAL	STAGE 2: CHECKER SANCTION	STAGE 3: CBS DISBURSEMENT
Field & Loan Officers	Branch Managers	Core Banking Push
• Zero Manual Data Entry: Land, crop, and applicant details auto-prefilled. • Scale of Finance Engine: Instant credit limit calculation based on crop acreage. • Real-Time Credit Bureau: 1-click CIBIL / Experian pull embedded in workflow. • Geo-Tagged Verification: Field officers upload verified crop photos. • Auto Appraisal Note: Standardized risk assessment draft in 60 seconds.	• Live SLA Countdown: Real-time timer preventing 7-day regulatory breach. • Anomaly Alerts: Flags multi-bank exposures or encumbrance conflicts. • 1-Click Sanction Letter: Bilingual statutory sanction memo generated. • Mandatory Rejection Codes: Rejections require structured rationale. • Transparent Recourse: Farmer receives instant SMS notification.	• Direct CBS Integration: Finacle, BaNCS, Flexcube via ISO-8583 / REST. • Digital Lien Creation: Charge registration communicated to Bhoomi. • Subvention Tracking: Automated interest subvention registry active. • e-Parihara Alignment: Linked to state disaster compensation registry. • Automated SLBC Filing: Eliminates manual branch reporting.

Unit Economics: Direct Bank Operational Savings

Demonstrating significant operational savings for participating banks alongside an asset-light, success-based fee structure.

Cost Component	Physical Paper Process	KisanKavach Digital	Operational Impact
Document Collection & Verification	Rs. 450	Rs. 0	100% Digital via Bhoomi RTC
Land & Crop Field Survey	Rs. 650	Rs. 250	62% Cost Reduction / Geo-Photo
Data Entry & Maker Appraisal	Rs. 400	Rs. 50	Auto-Prefilled from APIs
Branch Overhead & Storage	Rs. 700	Rs. 400	Zero Paper Storage / Audit-Ready
TOTAL COST PER APPLICATION	Rs. 2,200	Rs. 700	Rs. 1,500 Saved per Loan (68%)

SCALE ECONOMICS: A commercial bank processing 2,00,000 KCC applications annually saves Rs. 30.00 Crore in direct operating overhead each year.
COMMERCIAL MODEL: 0.35% origination fee on disbursed credit (Zero upfront Capex for banks) + Rs. 35,000 annual branch SaaS license.

State Command Center & Real-Time SLBC Oversight

Empowering the SLBC Convener, RBI, NABARD, and State Leadership with live operational visibility.

EXECUTIVE DRILLDOWNS	BANK PERFORMANCE SCORECARDS	SLA ENFORCEMENT ENGINE
District & Taluk Analytics	Branch Benchmarking	Collector Auto-Escalation
- Real-time heatmaps tracking credit velocity across all 31 districts & 240 taluks. - Compare disbursement pace against State Credit Plan targets. - Instant detection of regional credit gaps and underserved taluks. 1-click export of SLBC review reports in standard PDF and Excel formats.	- Live scorecards comparing SBI, Canara Bank, KVGB, Karnataka Bank, and DCCBs. - Turnaround time (TAT) tracking identifying branches exceeding 7 days. - Rejection taxonomy analysis highlighting branches with abnormal rejection spikes. - Officer productivity and field verification volume benchmarking.	- Real-time countdown timer on every loan application. - Automated escalation to District Collectors when SLA is breached. - Integrated citizen grievance system resolving complaints under 10 days. - Full statutory audit trail preventing arbitrary rejections.

CyberKavach: First-of-its-Kind Agrarian Cyber Defense

Protecting farmers from rising predatory loan apps, SMS phishing, and WhatsApp OTP financial scams.

CYBER THREAT VECTORS IN RURAL INDIA	CYBERKAVACH DEFENSIVE CAPABILITIES
• Fake PM-KISAN Links: Phishing SMS messages luring farmers with fake bonus credits to capture Aadhaar and bank credentials. • Predatory Loan APKs: Unregistered instant loan apps charging extortionate fees and siphoning contact lists. • OTP Social Engineering: Fraudsters posing as bank branch managers asking for debit card and OTP details. • Disbursement Draining: Siphoning freshly disbursed KCC loan funds before the farmer can purchase farm inputs.	✓ DigiKavach-Aligned Heuristics: Real-time ML and regex detection of fraudulent URLs and scam syntax. ✓ Bilingual Scanner (Kannada & English): Farmers paste suspicious messages into the app for instant threat analysis. ✓ Instant Plain-Language Warning: 'ALERT: Do not share OTP. This link will siphon your bank funds.' ✓ Bank Account Shield: Ensures disbursed agricultural credit stays protected within legitimate bank accounts.

Data Governance, Security & DPDP Act Compliance

Strict compliance with Indian data privacy mandates, RBI master directions, and banking cybersecurity norms.

DPDP ACT 2023 COMPLIANCE	BANK-GRADE CYBERSECURITY
• Explicit Notice & Consent: Farmer provides explicit digital consent prior to fetching Bhoomi or FRUITS records. • Revocable Permissions: Farmers can review and revoke consent records at any time via mobile. • Audit Consent Trail: Timestamped consent tokens logged with UIDAI / CeG provenance. • Zero Data Monetization: Strict prohibition on third-party commercial profiling.	• Encryption Standards: 256-bit TLS encryption in transit and AES-256 encryption at rest. • CERT-In Empaneled Audit: Platform primed for formal CERT-In vulnerability assessment. • Role-Based Access Control (RBAC): Separation of duties between Maker, Checker, and Admin. • Sovereign Hosting: 100% data residency within MeitY-empaneled Indian cloud data centers.

Statewide Rollout Roadmap & CBS Integration Plan

A structured, phased progression from verified 10-district pilot to statewide Karnataka ubiquity.

PHASE 1 (COMPLETED)	PHASE 2 (CURRENT)	PHASE 3 (SCALE)	PHASE 4 (NATIONAL)
10 Pilot Districts (2026)	All 31 KA Districts (2027)	Insurance & FPOs (2028-29)	AgriStack Federation (2030+)
• 10 Pilot districts live • Bhoomi/FRUITS sandboxes • 1,50,000 farmers registered • 1,05,000 KCC sanctions • Rs. 1,890 Cr credit flow • Zero ghost borrower cases	• Full rollout to 240+ taluks • CBS ISO-8583 push • Automated CIBIL pull • 5,00,000 active farmers • Rs. 6,750 Cr credit flow • 31 Collectorate desks	• PMFBY weather index claims • KSNDMC weather advisory • FPO group credit module • Input credit checkout APIs • Voice AI in local Kannada • 12,00,000 active farmers	• National AgriStack UFSP • Multi-state registry adaptors • e-NWR warehouse financing • 50,00,000 active farmers • Rs. 72,000 Cr credit flow • Sovereign DPI export

Proposed Resolutions for SLBC Plenary Approval

Actionable resolutions submitted for consideration and adoption by the State Level Bankers' Committee.

RESOLUTION 1: DIGITAL GATEWAY EMPANELMENT • Formal adoption of KisanKavach as the preferred digital lending orchestration gateway for KCC credit origination across Karnataka. • Recognition of digital Bhoomi RTC and FRUITS crop provenance as valid prefill for loan applications. • Standardizing the 7-day SLA countdown across all member bank branches.	RESOLUTION 2: CORE BANKING API CONNECTIVITY • Authorize technical integration between KisanKavach and member bank CBS gateways (Finacle, BaNCS, Flexcube). • Enable automated digital lien registration / charge creation communication with Bhoomi. • Provide sandbox access for CIBIL / credit bureau real-time pull integration.
RESOLUTION 3: JOINT IMPLEMENTATION TASKFORCE • Form a joint 5-member steering committee comprising: - SLBC Convener Representative (Canara Bank) - State Dept of Agriculture & Bhoomi Nodal Officers - Lead District Managers (LDMs) of Pilot Districts - Lead System Architect (Sarath Babu Rayaprolu)	RESOLUTION 4: STATEWIDE EXPANSION MANDATE • Approve statewide expansion across all 31 Karnataka districts. • Set target of onboarding 5,00,000 smallholder farmers in FY 2026-27. • Facilitate Rs. 6,750 Crore in prioritized agricultural credit through partner banks. • Mandate quarterly progress reviews at future SLBC plenary meetings.



Empowering Karnataka's Agrarian Economy

Rapid, Transparent & Fraud-Free Digital Agricultural Credit

Author & Presenter:	Sarath Babu Rayaprolu (Founder & Lead System Architect)
Production Web Platform:	https://kisan.digikavach.net
Live Demonstration:	Bank Maker-Checker Portal & District Command Center
Mobile Application:	Native Android APK Ready for Field Demonstration
Operations & Head Office:	Bengaluru, Karnataka Email: sarath@digikavach.net